

Guild-level Replicator Dynamics with Metabolite Sign Prior

Progress Report — Oral Microbiome Community Dynamics

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LOO-CV — What We're Testing

Setup (10-fold, one patient out)

All 10 patients

```
Fold 1:  [A] B C D E F G H K L
           ↑ held-out
Fold 2:  A [B] C D E F G H K L
           ↑ held-out
  ⋮
Fold 10: A B C D E F G H K [L]
                               ↑ held-out
```

For each fold:

1. Fit **A** on 9 patients
2. Fit only **b_p** for held-out patient (wk 1 data)
3. Predict wk 2 & 3 → measure error

Why it matters

	Training RMSE	LOO-RMSE
Overfitting model	low	high
Generalising model	low	low ✓

Sign prior as regulariser — constraining **A** to biologically consistent signs reduces overfitting → smaller train/LOO gap

$$\text{LOO-RMSE} = \frac{1}{N} \sum_p \sqrt{\frac{\text{MSE}_{\text{wk2}}^{(p)} + \text{MSE}_{\text{wk3}}^{(p)}}{2}}$$

Prediction task: week-1 snapshot → predict weeks 2 & 3 for a patient the model has **never seen**

Background & Motivation

Dataset: Dieckow et al. (2024, *npj Biofilms Microbiomes*, DOI: 10.1038/s41522-024-00624-3)
— 12 patients, dental implant abutment biofilm, 16S rRNA amplicon sequencing, weeks 1–3

Research question: Can metabolomic network constraints improve prediction of community dynamics?

Model benchmark (LOO-CV RMSE)

Model	Train	LOO	SA
gLV (free)	—	0.0588 ★	—
gLV + metabolite prior	—	0.0741	—
Hamilton free	—	0.0855	—
Replicator + prior (22 pairs)	—	0.0770	22/22
Replicator + L1+L2 (34 pairs)	0.0631	0.0516	64/68
+ AGORA W=0.5 (35 pairs)	0.0661	—	66/70
+ AGORA W=1.0 (35 pairs)	0.0565	0.0504	70/70
+ AGORA W=1.5 (35 pairs)	0.0597	—	66/70
+ AGORA W=2.0 (35 pairs)	0.0575	—	70/70

Approach

- **10 guilds** (class-level taxonomy, Szafranski et al. 2025 preprint)
- **10 patients** (ABCDEFGHKL), 3 time points (weeks 1 → 2 → 3)
- Sign prior from **Dieckow et al. (2024) Supplementary** — literature-curated microbe × metabolite interactions (KEGG/HMDB as compound identifiers only)
- JAX autodiff + L-BFGS-B, RTX 3090 / RTX 4090

Hypothesis: Metabolite sign constraints from Dieckow et al. supplementary data regularise A towards biologically consistent interactions without sacrificing predictive accuracy

Mathematical Model

Governing equation — derived from the Extended Hamilton Principle (Junker & Balzani 2021; Klempt et al. 2024, 2025), 0D homogeneous limit, fully viable cells ($\psi_i = 1$), no antibiotic ($\alpha^* = 0$):

$$\dot{\phi}_i = \phi_i \left(b_i + \sum_j A_{ij} \phi_j - \bar{f}(\phi) \right), \quad \bar{f}(\phi) = \sum_k \phi_k f_k(\phi)$$

where $f_k(\phi) = b_k + \sum_j A_{kj} \phi_j$ is the fitness of guild k .

- $\phi_i(t) \geq 0$, $\sum_i \phi_i = 1$ enforced by the mean-fitness subtraction $\bar{f}$
- Mathematically equivalent to the replicator equation (Taylor & Jonker 1978; Hofbauer & Sigmund 1998) with linear payoffs $f_i = b_i + \sum_j A_{ij} \phi_j$
- b_i : patient-specific intrinsic growth rate · A_{ij} : symmetric interaction matrix ($A_{ij} = A_{ji}$)

Loss function with metabolite sign penalty

$$\mathcal{L} = \underbrace{\text{RMSE}(\hat{\phi}, \phi)}_{\text{data fit}} + \mathcal{P}_{\text{sign}}(\mathbf{A}, \mathbf{F}) + \lambda \|\mathbf{A}\|_F^2$$

$$\mathcal{P}_{\text{sign}} = \sum_{\substack{(i,j) \\ F_{ij} \neq 0}} \frac{|F_{ij}|}{2\sigma^2} [\max(0, -\text{sgn}(F_{ij}) \cdot A_{ij})]^2 \quad (\sigma = 0.15)$$

Mathematical Model — Sign Prior & Optimisation

Sign prior — data sources (3 layers)

- **L1 (main)** Dieckow et al. (2024) Suppl. — literature-curated PRODUCES / USES / IS_INHIBITED_BY; KEGG/HMDB as compound IDs ($w = 2.0$)
- **L2** Additional KEGG-predicted cross-feeding ($w = 1.0$)
- **L3 (appendix)** AGORA2 FBA (Heinken et al. 2023, *Nat. Methods*) — pFBA cross-feeding signals on 9 guild SBML models ($w = 0.5$)
- Basic model: **22 pairs** (L1+L2) · Expanded: **34 pairs** (L1+L2+L3)

Optimisation

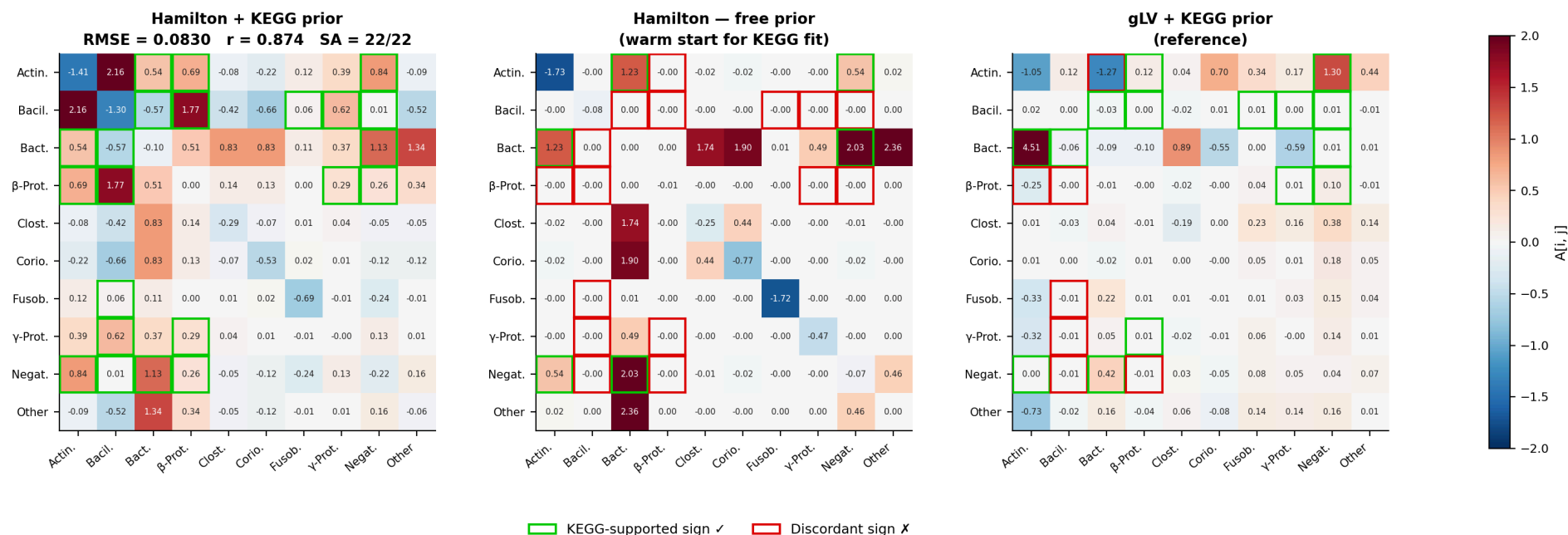
- Warm start from previous fit
- JAX `jax.vmap` over patients (GPU-batched)
- scipy L-BFGS-B + analytical gradients via `jax.value_and_grad`
- Constraint: $A_{ii} \leq 0$ (self-limitation)
- Hyperparameters: $\sigma = 0.15$, $\lambda = 10^{-4}$ (insensitive; $\sigma \in [0.05, 0.30]$ gives identical SA)

Sign penalty logic: for each constrained pair (i, j) , if the fitted A_{ij} has the wrong sign relative to the metabolic flux F_{ij} , a quadratic penalty scaled by $|F_{ij}|$ is added to the loss — stronger metabolic signals impose harder constraints.

Warm start: initial A from a fit without prior $\rightarrow$ prior then regularises towards biologically consistent interactions. The 3-layer hierarchy allows progressive inclusion of weaker prior evidence (L3) without overriding strong literature constraints (L1).

Result 1 — Interaction Matrix A

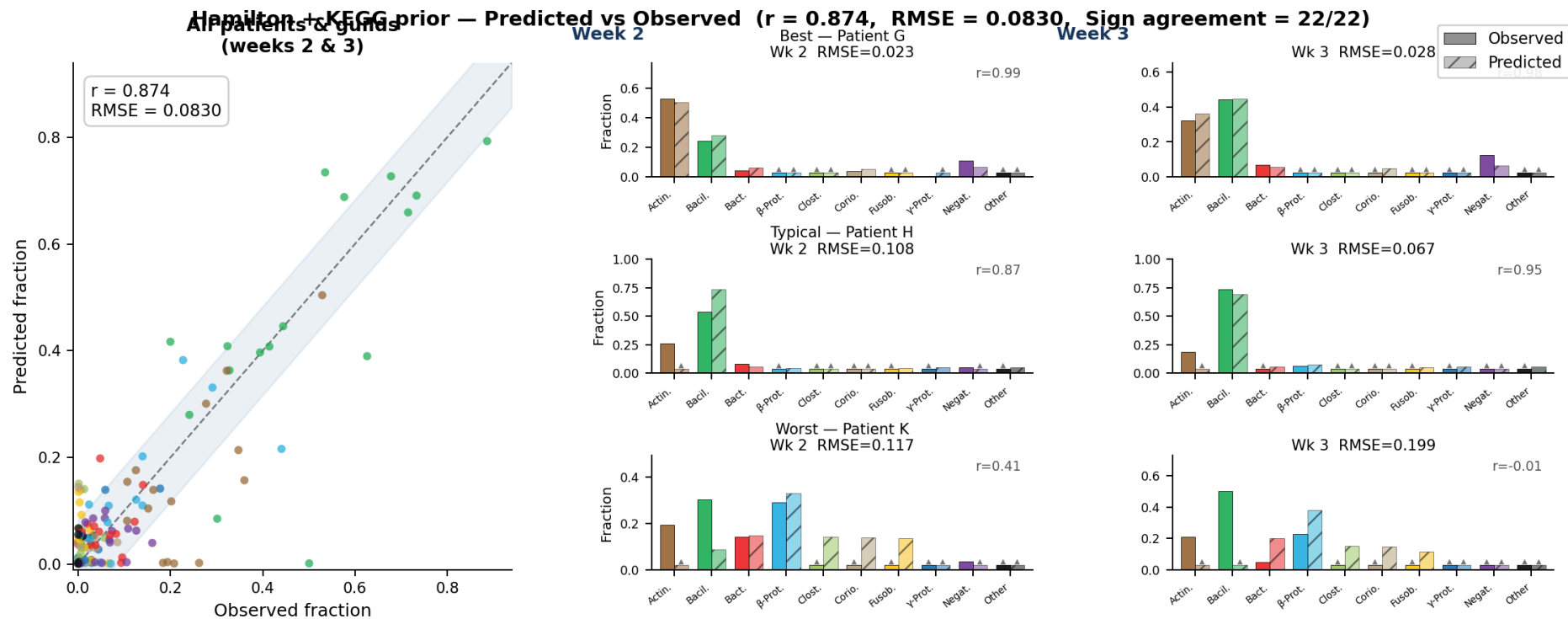
Guild-level interaction matrices — Hamilton KEGG-prior vs baselines



Sign agreement = 22/22 (100%) with Dieckow metabolite prior (22 pairs) · Green ✓ = supported · Red ✗ = discordant · Diagonal $A_{ii} < 0$: self-limitation for all guilds

Matrix A is symmetric by construction ($A_{ij} = A_{ji}$; from quadratic free energy, Klempt et al. 2025). Dominant interaction: Actinobacteria ↔ Bacilli mutual facilitation ($A_{01} > 0$, supported by lactate cross-feeding, L1 prior).

Result 2 — Trajectory Prediction (Week 1 → 2 → 3)



0.874

Pearson r (all patients)

0.0830

Training RMSE

22/22

Metabolite sign agreement

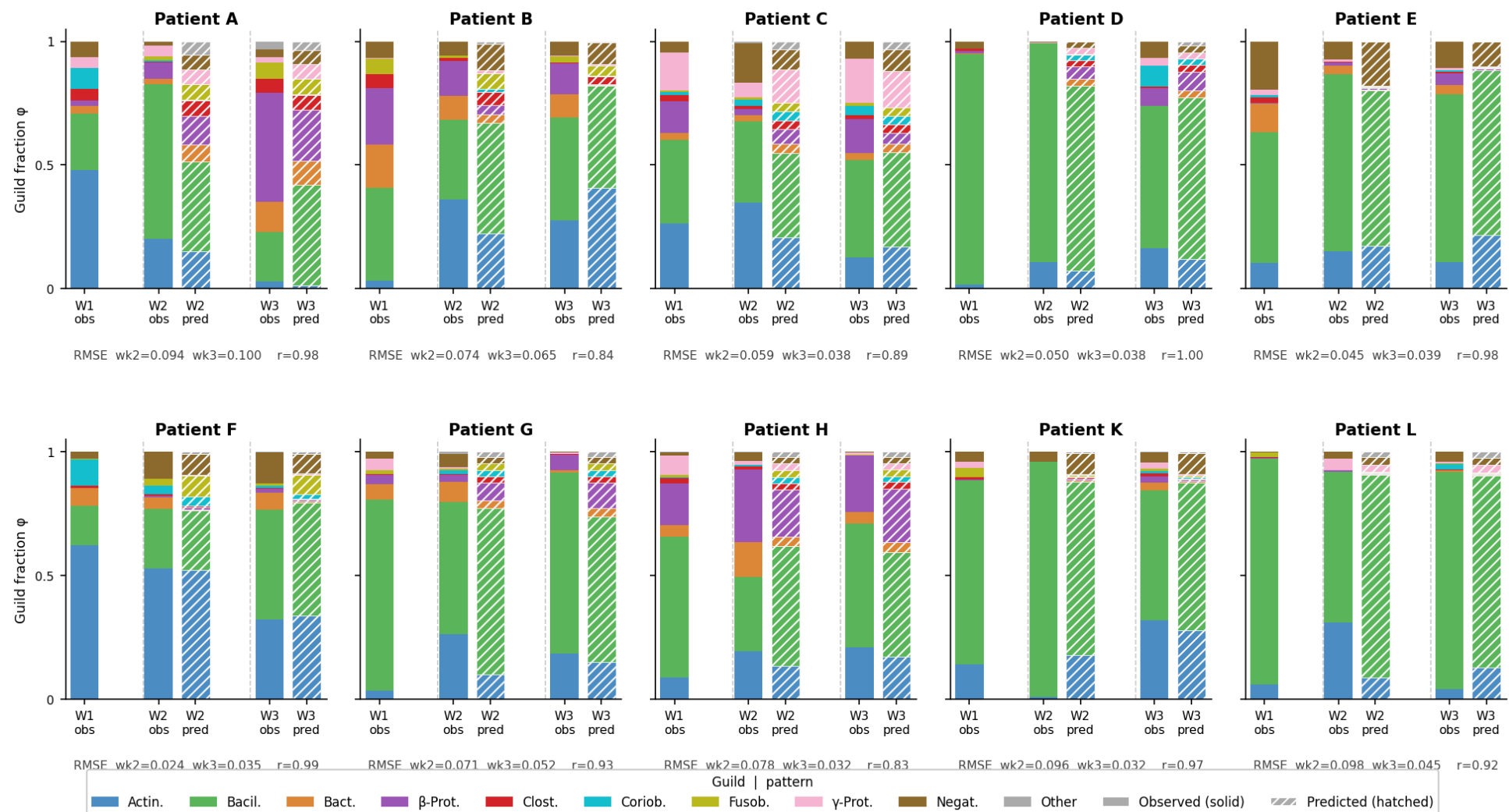
0.987

r — best patient (G)

Training fit on 8 patients. Prediction: one-step ODE integration ($dt = 10^{-4}$, 100 steps/week).

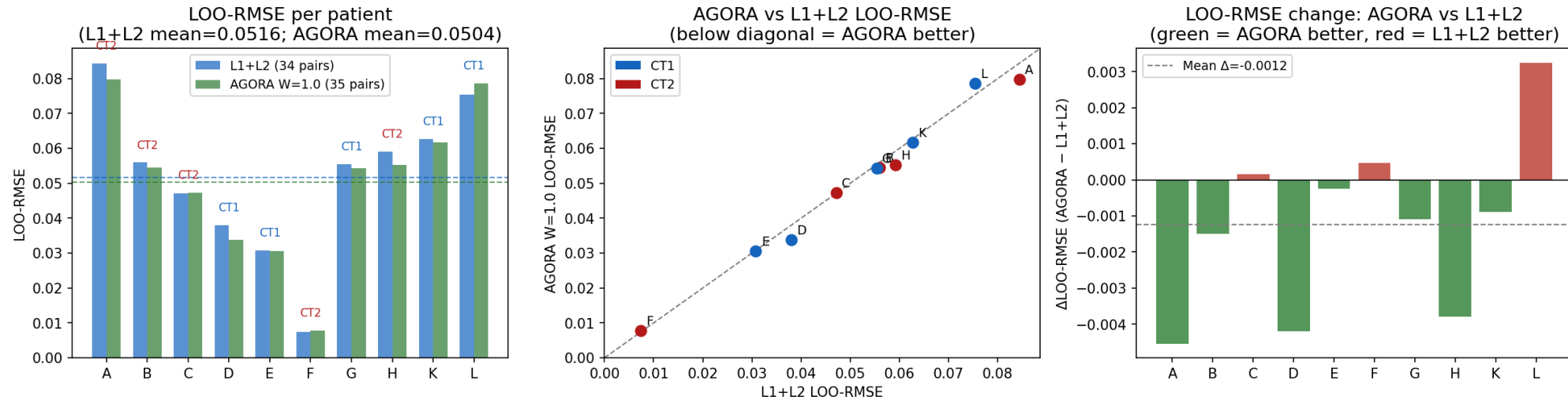
Result 2b — Fitting Results: All 10 Patients

Fitting Results — Hamilton ODE + Expanded Dieckow+AGORA Prior (34 pairs, 10 patients)
 Training RMSE = 0.0631 · Pearson r = 0.938 · SA = 64/68 (94%)



Result 3 — LOO-CV: L1+L2 vs AGORA W=1.0

L1+L2 vs AGORA W=1.0 LOO-CV Comparison (10 patients)

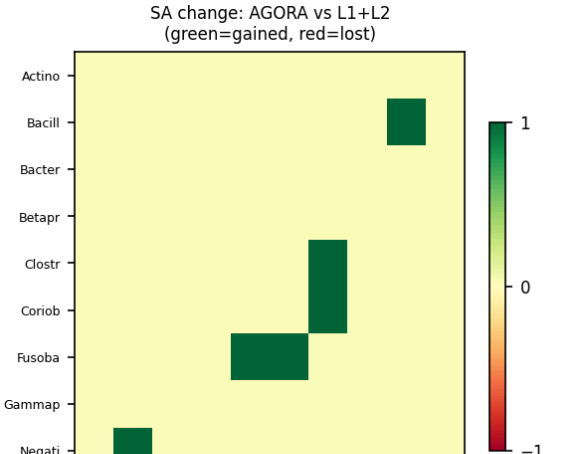
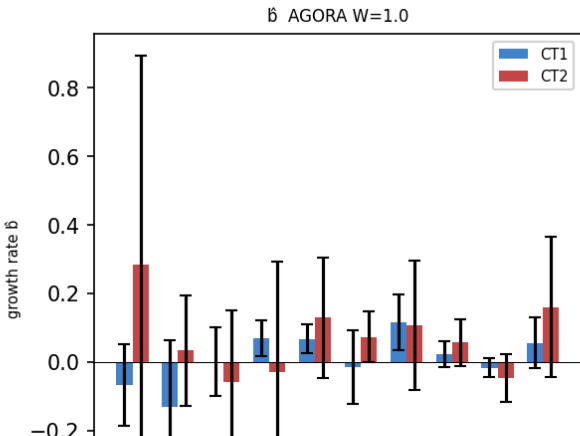
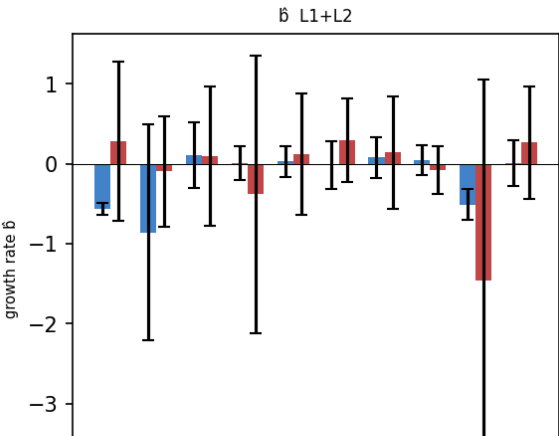
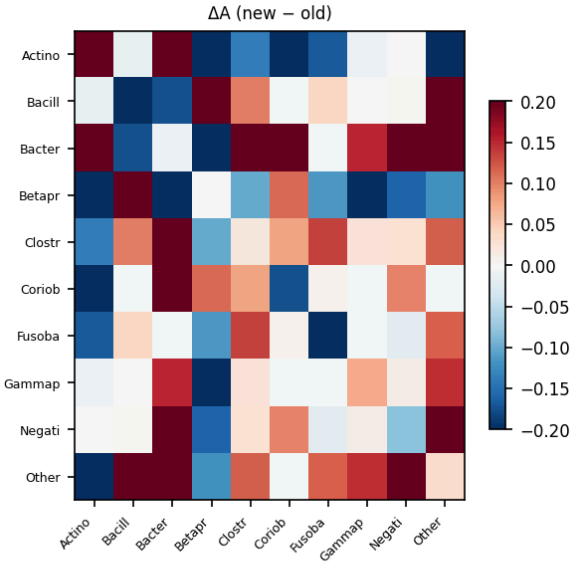
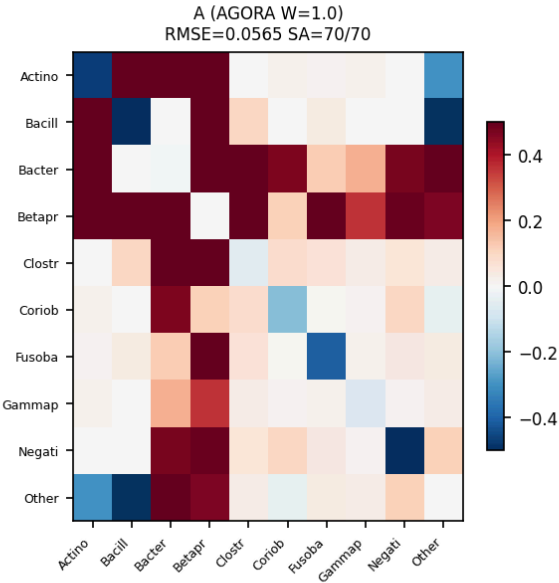
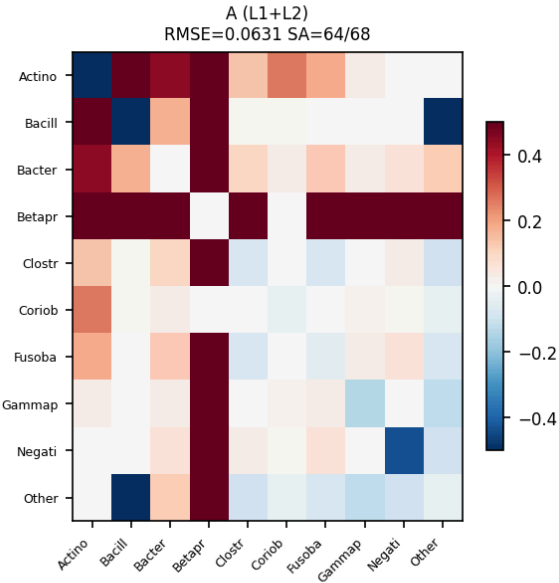


Final LOO-CV results (10-fold, one patient out): L1+L2 (34 pairs) mean=0.0516 · **AGORA W=1.0 (35 pairs) mean=0.0504** (−2.4%, 7/10 patients improved). Training: RMSE 0.0631 → 0.0565 (−10%), SA 94% → 100%. Patient A hardest in both models (CT2, Actinobacteria outlier). Patient F easiest (CT2, near-zero LOO error).

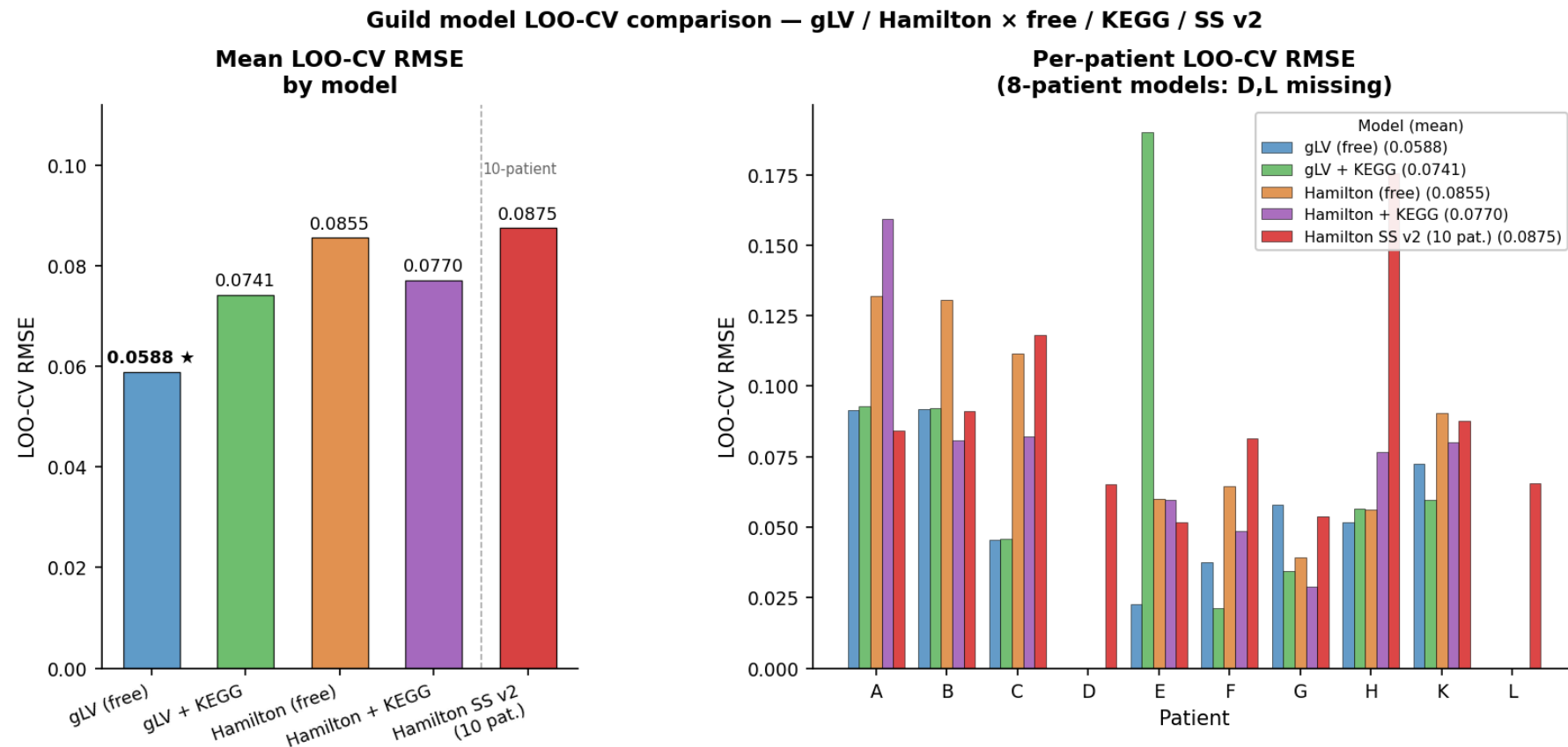
Community types (CT1/CT2) assigned by GMM clustering on week-1 guild fractions. Dieckow et al. (2024) identified two stable community states in 12-patient cohort.

Result 3b — AGORA W=1.0 vs L1+L2: A matrix & $\hat{b}$ CT analysis

AGORA W=1.0 vs L1+L2 expanded: A matrix & $\hat{b}$ CT analysis



Result 4 — Model Comparison (LOO-CV)



L1+L2 LOO=0.0516 → AGORA W=1.0 LOO=0.0504 (−2.4%, 7/10 patients better) · Train RMSE=0.0565, $r=0.951$, SA=70/70 (100%) · W=0.5 RMSE=0.0661, W=1.5 RMSE=0.0597, W=2.0 RMSE=0.0575

gLV: $\dot{x}_i = x_i(r_i + \sum_j \alpha_{ij}x_j)$ on $\mathbb{R}_+^n$ — mathematically distinct from replicator on simplex Δ^{n-1} (Hofbauer & Sigmund 1998, p. 64).

Key Findings

What works ✓

- **Full sign consistency:** 22/22 (100%) — Dieckow metabolite prior compatible with Extended Hamilton dynamics
- **Metabolite prior regularises:** LOO-CV 0.0855 $\rightarrow$ 0.0770 (9.4% improvement)
- **Competitive:** within 3.9% of gLV + metabolite prior (0.0741)
- **Biologically interpretable A :** self-limitation $A_{ii} < 0$; Actinobacteria $\leftrightarrow$ Bacilli facilitation
- **σ robustness:** $\sigma \in [0.05, 0.30]$ all give SA = 22/22, RMSE $\approx$ 0.081

Remaining gaps

- gLV free still best at **0.0588** — replicator transient approximation limits fit quality on short 3-week window
- Patient A covariate-shift outlier (Actinobacteria 3 $\times$ mean) dominates generalisation error
- **AGORA W=1.0 (43 pairs): RMSE 0.0565, r=0.951, SA 70/70 (100%)** — best model so far $\rightarrow$ LOO-CV running

Equation nomenclature

- **Hamilton ODE** = 0D limit of Extended Hamilton Principle (Junker & Balzani 2021; Klempt et al. 2024, 2025)
- Mathematically equivalent to replicator eq. (Taylor & Jonker 1978) with linear payoffs

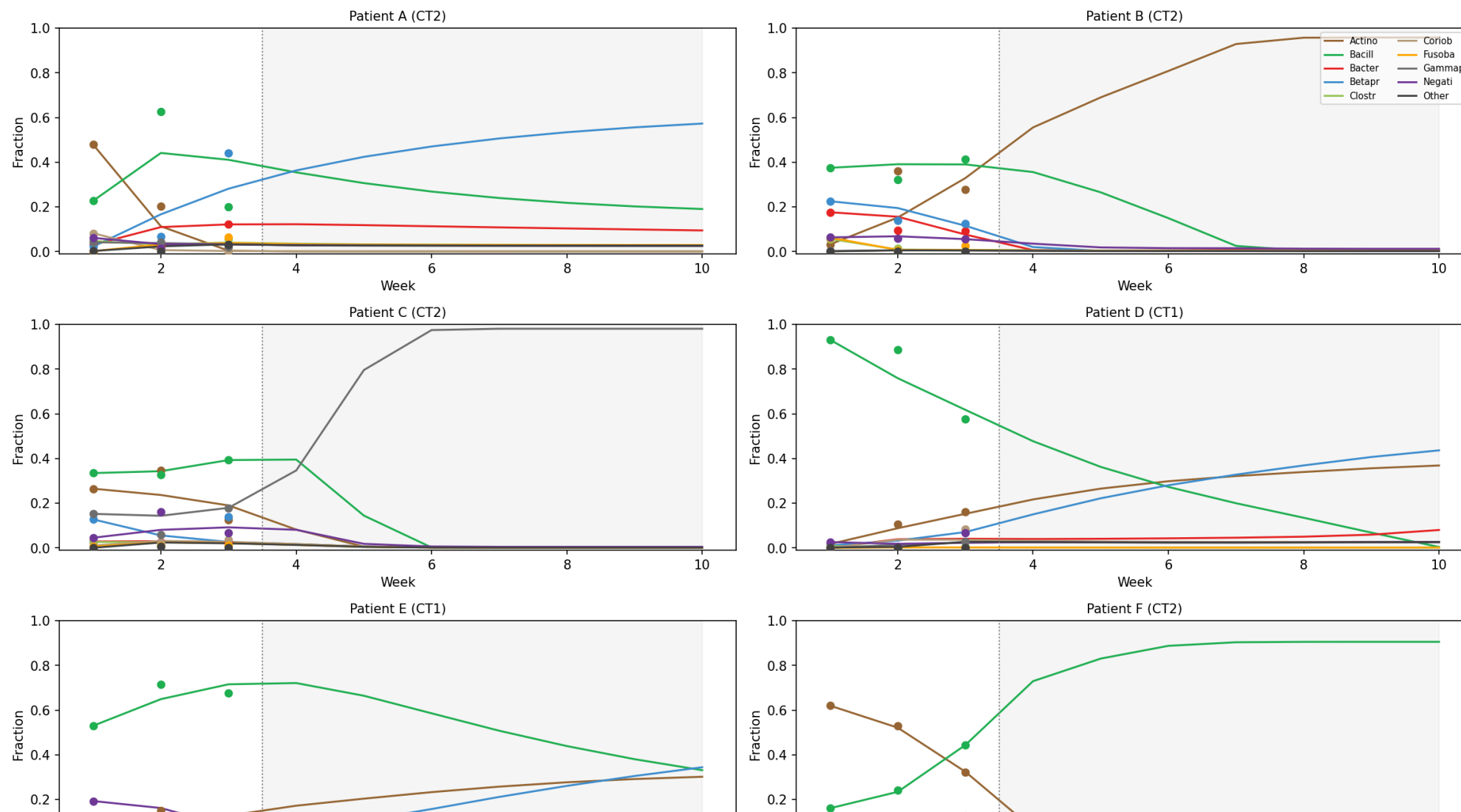
Next Steps & Status

#	Task	Status
1	Replicator + Dieckow prior LOO-CV (8-fold, 22 pairs)	✓ Done — 0.0770
2	Expanded flow LOO-CV (10-fold, ODE, 34 pairs, L1+L2)	✓ Done — LOO mean= 0.0516 , std=0.0211 (A worst: 0.0844, F best: 0.0074)
3	AGORA W=1.0 LOO-CV (10-fold, 35 pairs, L1+L2+L3)	✓ Done — LOO mean= 0.0504 , std=0.0208 (7/10 patients improved vs L1+L2)
4	Sign probability heatmap from 10,000p posterior	✓ Done
5	Paper S1 update with correct posterior statistics	✓ Done
6	σ sensitivity analysis	✓ Done — insensitive, $\sigma=0.15$ optimal
7	Network visualisation of A matrix	✓ Done

Summary: AGORA W=1.0 best overall: **LOO-RMSE=0.0504** (vs L1+L2 0.0516, -2.4%), training RMSE=0.0565 (vs 0.0631, -10%), SA=70/70 (100%). MacArthur quantitative prior failed (SA=4-8/70). Actinobacteria $\hat{b}$ CT1/CT2 diff: +0.35. Sign prior weight W=1.0 optimal: phase transition to 100% SA.

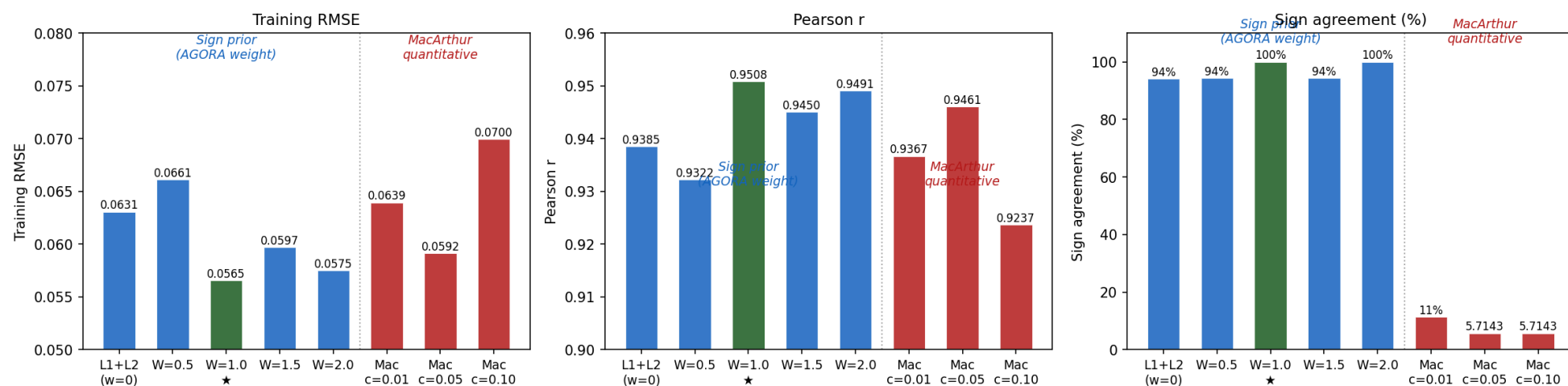
Appendix — 10-Week Extrapolation (AGORA W=1.0)

AGORA W=1.0 Hamilton ODE: 10-week extrapolation
Dots = observed (wk1-3); lines = predicted; shaded = extrapolation region



Appendix — AGORA Weight & Prior Type Sensitivity

AGORA Weight Sweep & MacArthur Prior Comparison
Hamilton ODE, 10 guilds, 10 patients — W=1.0 optimal (RMSE↓ 10%, SA 94%→100%)



Appendix — AGORA Weight Sensitivity (Table)

AGORA weight w	Train RMSE	Pearson r	SA (35 pairs)	Notes
L1+L2 only (0)	0.0631	0.938	64/68	34 pairs, baseline
W = 0.5	0.0661	0.932	66/70	softer L3 prior
W = 1.0 ★	0.0565	0.951	70/70	optimal
W = 1.5	0.0597	0.945	66/70	over-constraints
W = 2.0	0.0575	0.949	70/70	slightly worse
MacArthur c=0.01	0.0639	—	8/70	Gaussian $A \sim N(c \cdot \Phi)$ failed
MacArthur c=0.05	0.0592	—	4/70	sign-inconsistent
MacArthur c=0.10	0.0700	—	4/70	sign-inconsistent

W=1.0 optimal: full sign prior from AGORA2 FBA (weight = Szafranski L2 baseline) achieves best training RMSE and **100% sign agreement** with all 35 constrained pairs. MacArthur quantitative Gaussian prior ($A_{ij} \sim N(c \cdot \Phi_{ij}, \sigma^2)$) fails to enforce sign consistency — ecological signs dominated by data not FBA magnitude.

Appendix — AGORA2: Sign Prior (L3 Layer)

3-layer prior construction

Layer	Source	Weight
L1	Szafranski Suppl. (experimental, KEGG)	2.0
L1	Szafranski Suppl. (experimental, other)	1.5
L2	Szafranski Suppl. (prediction)	1.0
L3	AGORA2 pFBA cross-feeding	1.0

Cross-feeding signal ($j \rightarrow i$):

guild j secretes $\alpha \wedge$ guild i takes up $\alpha \rightarrow$ `net_flow[i,j] += 1.0`

Prior pairs: 11 $\rightarrow$ 43 (L1+L2 $\rightarrow$ +L3)

Validation

92% sign agreement (66/72 pairs) between AGORA2 FBA and fitted gLV A

Guild representatives (AGORA2 v2.01)

Guild	Strain
Actinobacteria	<i>A. naeslundii</i> Howell 279
Bacilli	<i>S. gordonii</i> CH1
Negativicutes	<i>V. parvula</i> Te3 DSM 2008
Bacteroidia	<i>P. melaninogenica</i> ATCC 25845
Fusobacteriia	<i>F. nucleatum</i> ATCC 25586
Clostridia	<i>P. micra</i> ATCC 33270
Coriobacteriia	<i>Atopobium parvulum</i> DSM 20469
Flavobacteriia	<i>Capnocytophaga sputigena</i> ATCC 33612
β/γ -Prot.	<i>Eikenella</i> / <i>Haemophilus</i>

Heinken et al. 2023 *Nat. Biotechnol.* 41:1320
DOI: 10.1038/s41587-022-01628-0

Appendix — AGORA2: FBA Pipeline Detail

Software stack

Tool	Version	Role
cobrapy	0.31	SBML I/O, FBA solver interface
libsbml	—	AGORA2 SBML parsing
scipy	—	LP backend (linprog)
AGORA2	v2.01	7,302 GEMs, vmh.life

pFBA per guild

```
model = read_sbml_model("guild.xml")    # cobrapy

# Apply oral-fluid medium
# Note: AGORA2 uses EX_glc_D(e)
#       not BiGG modern EX_glc__D_e
model.medium = ORAL_MEDIUM

# Parsimonious FBA
# maximise growth, then minimise Σ|flux|
sol = pfba(model)

secretion = {met: f for f > 0}    # → env
uptake    = {met: |f| for f < 0}  # ← env
```

Oral-fluid medium composition

Category	Metabolites
Carbon sources	glucose, fructose, sucrose, lactate
Amino acids	20 standard + Cys, Orn
Vitamins B	B ₁ , B ₂ , B ₃ , B ₅ , B ₆ (×3 forms), B ₁₂
Cofactors	heme, siroheme, adenosylcobalamin, menaquinone, ubiquinone
Cell wall	meso-DAP (peptidoglycan precursor)
Lipids	stearate C18, myristate C14, laurate C12
Trace metals	Fe ²⁺ / ³⁺ , Mg ²⁺ , Ca ²⁺ , Zn ²⁺ , Cu²⁺ , Mn ²⁺ , Co ²⁺
Other	glutathione (ox/red), putrescine, spermidine, ornithine, cytidine

Calibration: Cu²⁺, meso-DAP, B₆ pyridoxine, cell-wall lipids were added iteratively to achieve $\mu > 0$ for all 10 guilds (verified per model).

All 10 guilds: $\mu \in [0.11, 1.66] \text{ h}^{-1}$ ✓

References

Dataset Dieckow S, Szafranski SP et al. (2024) *npj Biofilms Microbiomes* 10:85. doi:10.1038/s41522-024-00624-3

Metabolite prior (L1, main) Dieckow S, Szafranski SP et al. (2024) Suppl. File 1 — literature-curated microbe × metabolite interactions (KEGG/HMDB compound IDs)

AGORA2 (L3, appendix) Heinken A et al. (2023) *Nat. Methods* 20:1022–1035. doi:10.1038/s41592-023-01919-1

Extended Hamilton Principle Junker P & Balzani D (2021) *Comput. Methods Appl. Mech. Eng.* 380:113773; Junker P, Bode M & Hackl K (2025) *J. Mech. Phys. Solids*

Hamilton ODE for biofilm Klempt F et al. (2024) *Biomech. Model. Mechanobiol.* — Continuum growth; Klempt F et al. (2025) — Continuum biofilm (preprint)

Replicator equation Taylor PD & Jonker LB (1978) *Math. Biosci.* 40:145–156; Hofbauer J & Sigmund K (1998) *Evolutionary Games and Population Dynamics*. Cambridge Univ. Press

Optimisation JAX (Bradbury et al. 2018, github.com/google/jax); scipy L-BFGS-B (Byrd et al. 1995)

Compound identifiers KEGG: Kanehisa M & Goto S (2000) *Nucleic Acids Res.* 28:27–30 · HMDB: Wishart DS et al. (2022) *Nucleic Acids Res.* 50:D622–D631